HS4002: Lecture 12b

LLMs, social science, and silicon sampling.

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LLMs: a primer

What an LLM is

- A model of what text comes next.

$$p(x_n \mid x_1, \ldots, x_{n-1})$$

- Autocomplete, at an enormous scale.

Tokens

Models read tokens, not words.

"Unbelievable" $\rightarrow$ [un] [believ] [able]

Context

Everything the model is given = its context.

"Can you come . . ."

"home" ✓ "purple" ✗

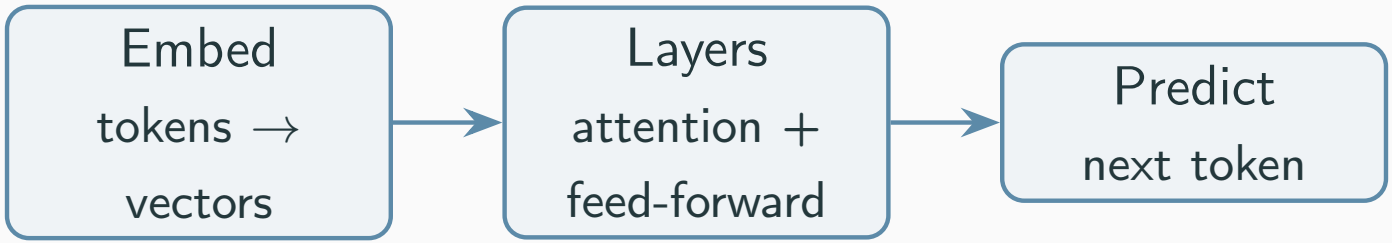
Attention

"sat by the river bank" vs. "a loan from the bank"

- Each token weighs every earlier token: *which ones matter here?*
- So every word of an answer is shaped by the whole context.
  - → Including a persona: "You are a 45-year-old Malay woman in Tampines . . ."
- Small changes in a prompt can shift the answer.

The transformer

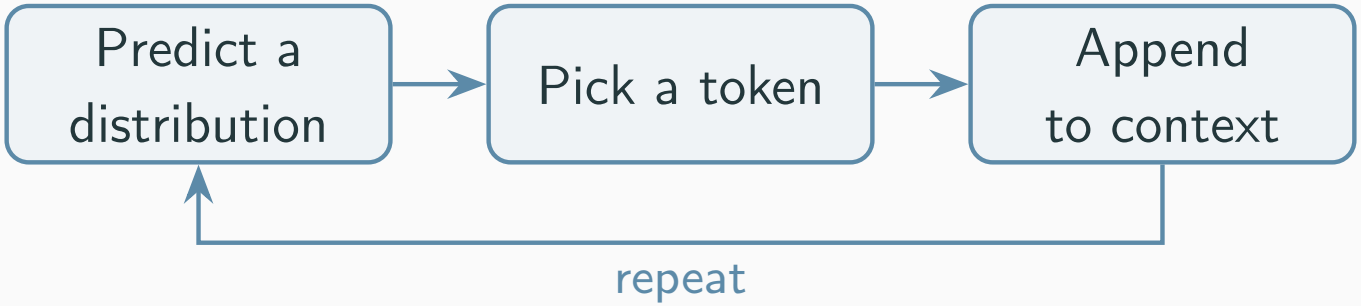
The design behind almost every LLM (Vaswani et al. 2017).



Parameters and scale

- Parameters: the numbers set during training.
- GPT-3 (2020):
  - $\rightarrow$ 175 billion parameters.
  - $\rightarrow$ 45+ terabytes of training text.
- Today's frontier models: larger, and mostly undisclosed.

Token generation



- One token at a time (*autoregressive*).
- It can't go back and revise.
- "Reasoning" models think first, then answer.

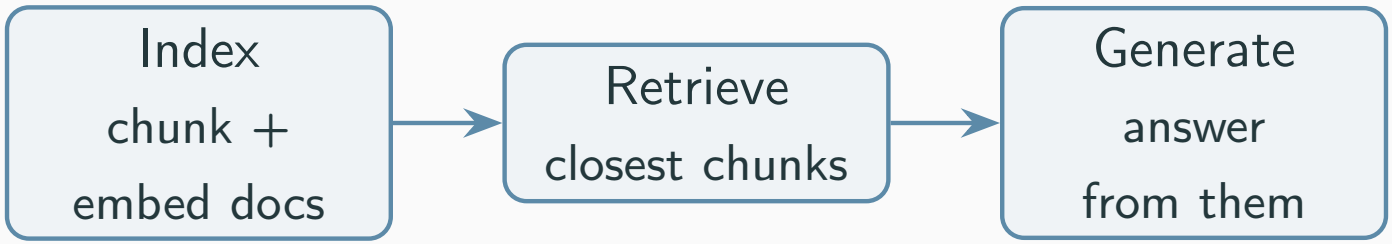
How tokens get picked

Usually a random draw, weighted by probability.

- Temperature
  - $\rightarrow$ Low $\rightarrow$ predictable.
  - $\rightarrow$ High $\rightarrow$ varied, then incoherent.
- Top-p: ignore the long tail of unlikely tokens.

Retrieval-augmented generation (RAG)

Models don't know your files, or today's news.



Find the relevant text first, then put it into the context.

RAG in practice

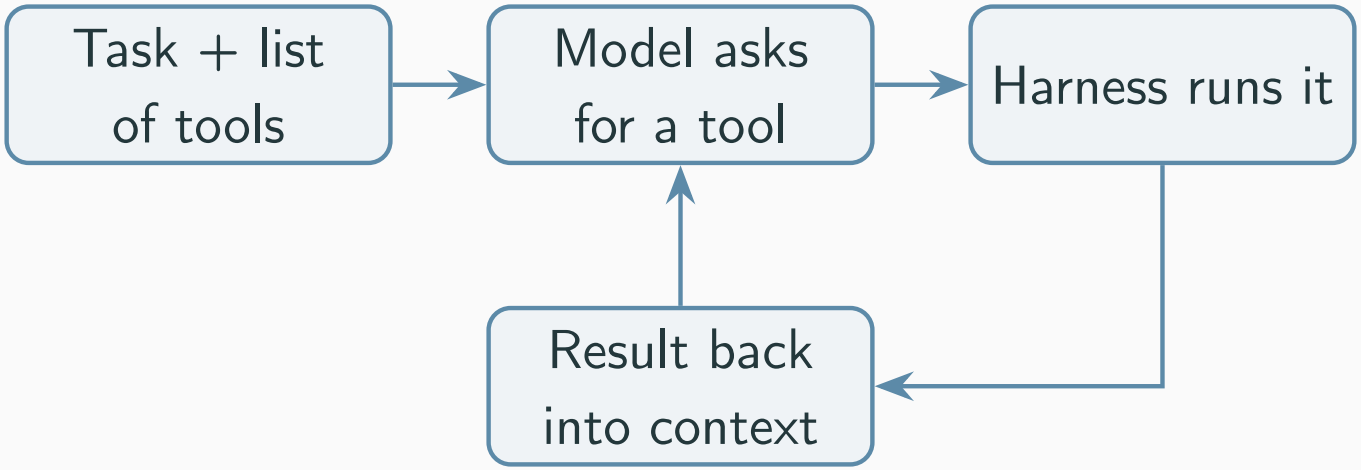
- You've probably used it:
  - → Chatbots searching the web.
  - → NotebookLM on your PDFs.
  - → Cursor finding files in your project.
- Adds to the context; doesn't retrain the model.
- Reduces hallucination, but doesn't remove it.
  - → Check the citations.

LLMs in coding tools

From autocomplete to agents

1. Autocomplete: suggests the next few lines.
2. Chat and inline edits: you ask, it proposes, you accept or reject.
3. Agents: given tools, works through many steps.

How agentics work



- Stops when the model answers without asking for a tool.
- "Using a tool" = generating text + a program acting on it.

Cursor

A code editor (built on VS Code) with LLMs throughout.

- Tab: predicts your next edit.
- Inline edits and chat: review changes as *diffs*.
- Agent mode: edits many files, runs commands.
- Choose models from several providers.
- Indexes your project to find relevant files.

Claude Code

Anthropic's coding agent, mainly in the terminal.

- No editor: describe the task in plain language.
- The agent loop does the rest:

  $\rightarrow$ search $\rightarrow$ read $\rightarrow$ edit $\rightarrow$ run $\rightarrow$ fix.
- Asks permission before edits and commands.

Silicon sampling

How do LLMs represent the social world?

- Do they "understand"? A live debate.
- More useful questions for us:
  - $\rightarrow$ How do they represent the social world?
  - $\rightarrow$ What do they carry over from their training text?

LLMs as research subjects

Instead of asking people,

ask a model to answer *as* them.

In economics: *homo silicus*

- GPT-3, given "endowments" (e.g. cares only about fairness).
- Re-runs classic studies:
  - → Dictator games.
  - → Price of snow shovels after a storm.
  - → Status quo bias.
- Broadly matched the human results.

In psychology

- Moral judgements: $r \approx 0.95$ with humans.
- Milgram: reproduced.
- Survey experiments: predicted effects, $r \approx 0.86$.
- Centaur: an LLM tuned on 160 experiments.

Why survey research?

Polling is under strain:

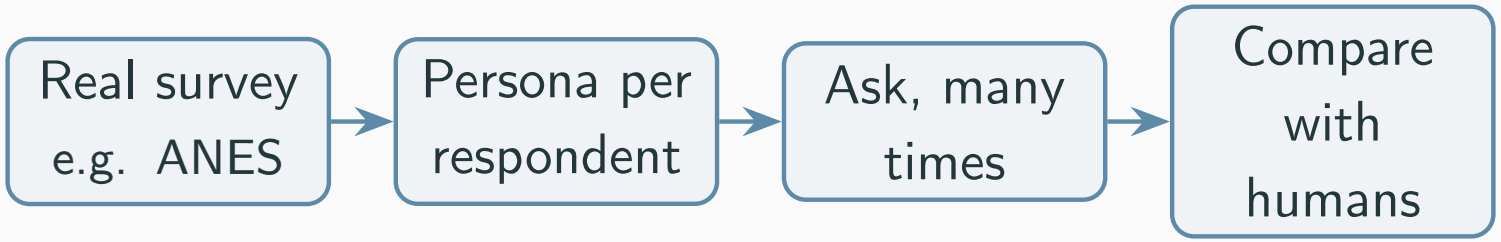
- Rising costs.
- Falling response rates.
- Worries about accuracy.

Cheap, fast opinion data would be very attractive.

Advantages of silicon sampling

- Cost and speed: thousands of answers in minutes.
- Hard-to-reach groups: no recruitment.
- Piloting: test designs before fieldwork.
- Sensitive topics: no one is put at risk.

How silicon sampling works



- Same make-up as the human sample, by construction.
- e.g. 30 answers $\times$ 7,530 ANES respondents.

An example persona

```
Racially, I am white. I am a man. Ideologically,
I describe myself as conservative. Politically,
I am a strong Republican. I attend church every
week. I am 52 years old. I am very interested in
politics. In 2016, I voted for
```

The model continues the text: Republican or Democrat?

Algorithmic fidelity

Algorithimic fidelity: How closely do silicon samples match samples from respondents?

The optimistic case

GPT-3 vs. the ANES:

- Romney 2012: 39.1% (silicon) vs. 40.4% (ANES).
- Vote correlations: 0.90, 0.92, 0.94 (2012–2020).
- Strong partisans: near-perfect.
- Weaker for some groups: e.g. 0.71 for Black respondents (2012).

The sceptical case

ChatGPT feeling thermometers vs. the ANES:

- ✓ Means and rankings.
- ✗ Variance: far too small.
- ✗ Subgroups: 10–20 points too extreme.
- ✗ Regressions: often disagree.
- ✗ Stability: shifts with wording, and over time.

Beyond the US: Germany

GPT-3.5 predicting 2017 votes (GLES):

- Too many Greens, Left, non-voters.
- Too few FDP and AfD.
- Fine for "typical" partisans; misses cross-pressured voters.

Synthetic panels

Silicon sampling has left the lab

Synthetic respondents are now a product.

Qualtrics' synthetic panels

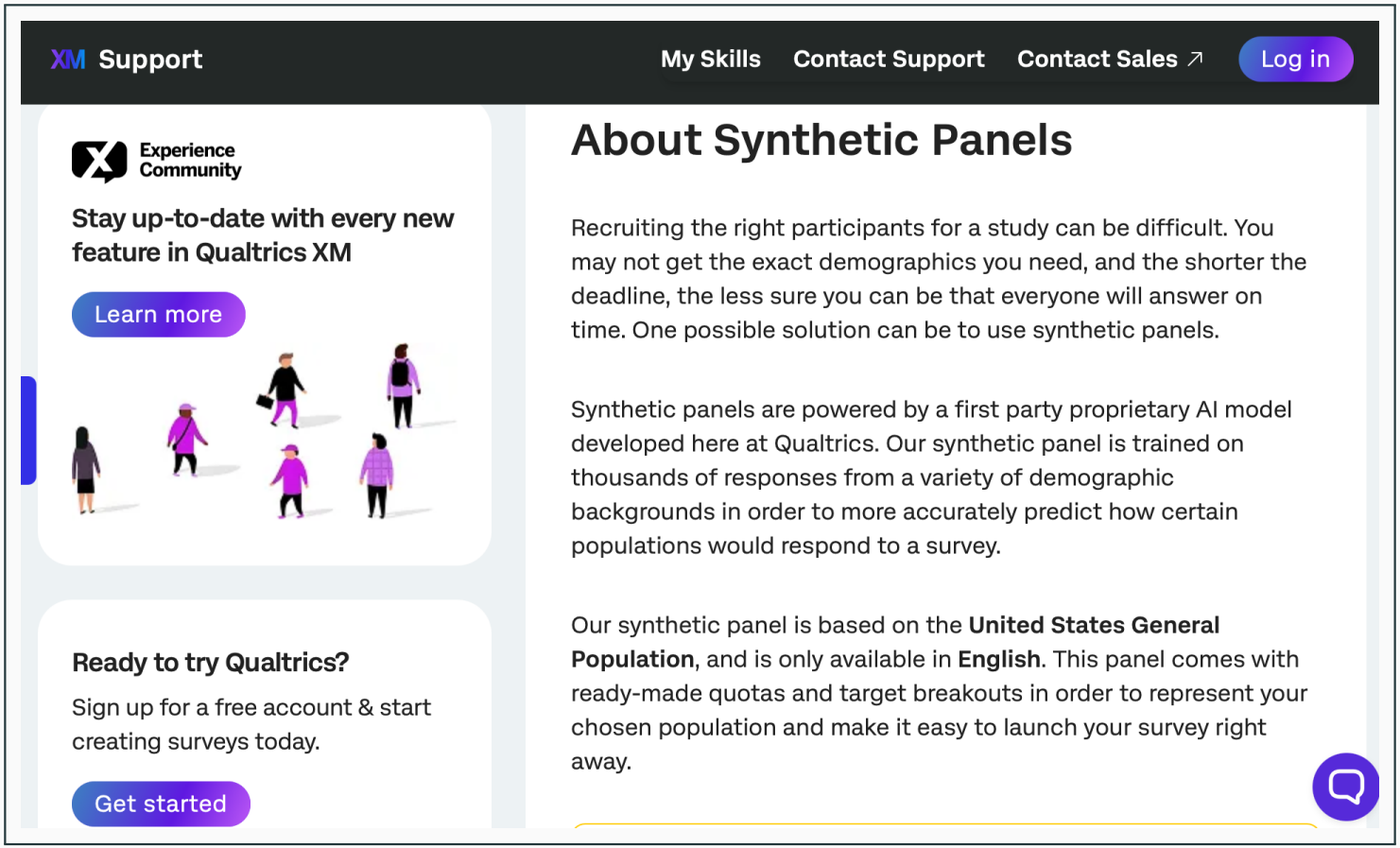


Figure 1: https://www.qualtrics.com/support/survey-platform/distributions-module/synthetic-panels/

Qualtrics' synthetic panels

- A "first party proprietary AI model".
- Order 50–10,000 synthetic responses, with quotas.
- US general population; English only.
- No images, videos or links.

Expected Parrot

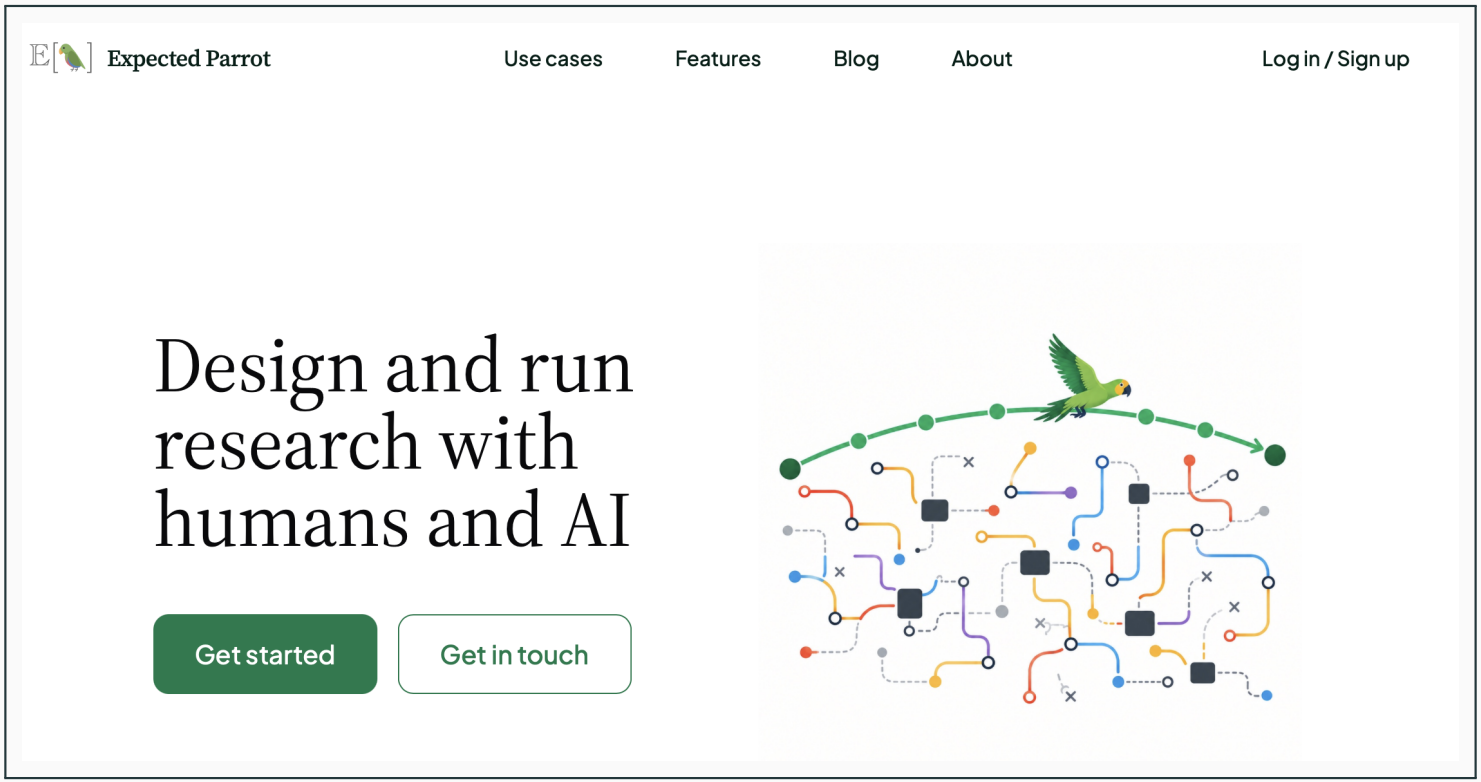


Figure 2: https://www.expectedparrot.com

End.